

DEMOTECH INC. — INTERNAL TECHNICAL GUIDE

Engineering Division | Last Updated: March 2025

SYSTEM ARCHITECTURE OVERVIEW

[illegible]

DemoTech's core platform runs on a microservices architecture deployed on AWS. The frontend is built in React 18 with TypeScript. The backend consists of 12 independent services communicating via REST and gRPC. All services are containerised using Docker and orchestrated through Kubernetes (EKS).

Primary database: PostgreSQL 15 (AWS RDS)

Cache layer: Redis 7.0 (AWS ElastiCache)

Message queue: Apache Kafka

File storage: AWS S3

CDN: CloudFront

API DOCUMENTATION

[illegible]

Base URL (Production): <https://api.demotech.io/v2>

Base URL (Staging): <https://staging-api.demotech.io/v2>

Authentication:

All API calls require a Bearer token in the Authorization header. Tokens expire after 24 hours. Refresh tokens are valid for 30 days. Token endpoint: POST /auth/refresh

Rate Limits:

Standard tier: 1,000 requests per minute

Premium tier: 10,000 requests per minute

Exceeded limits return HTTP 429 with Retry-After header.

DEPLOYMENT PROCESS

[illegible]

Step 1: Create a feature branch from main

Step 2: Submit pull request — minimum 2 approvals required

Step 3: CI pipeline runs (jest tests + ESLint + build check)

Step 4: Merge to staging branch triggers auto-deploy

Step 5: QA team runs smoke tests on staging (2 hours)

Step 6: Release manager approves production deploy

Step 7: Deploy runs between 2:00 AM – 4:00 AM IST only

Step 8: Rollback procedure: revert merge + redeploy

INCIDENT RESPONSE

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P0 (Critical — full outage):

Response time: 15 minutes

Escalation: On-call engineer → CTO within 30 min

War room: Slack #p0-incidents channel

P1 (Major — partial outage):

Response time: 1 hour

Escalation: On-call engineer → Engineering Manager

P2 (Minor — degraded performance):

Response time: 4 hours

Handled during business hours

TROUBLESHOOTING COMMON ISSUES

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Issue: API returns 503 unexpectedly

Fix: Check EKS pod health in AWS console.

Run: `kubectl get pods -n production`

If pods are CrashLooping, check logs:

`kubectl logs {pod-name} -n production`

Issue: Redis connection timeout

Fix: Check ElastiCache cluster status.

Verify security group allows port 6379.

Flush Redis if memory > 90%: `redis-cli FLUSHDB`

Issue: Build fails on CI pipeline

Fix: Ensure Node version matches `.nvmrc` (v20.11.0)

Delete `node_modules` and reinstall dependencies

Check for TypeScript strict mode violations